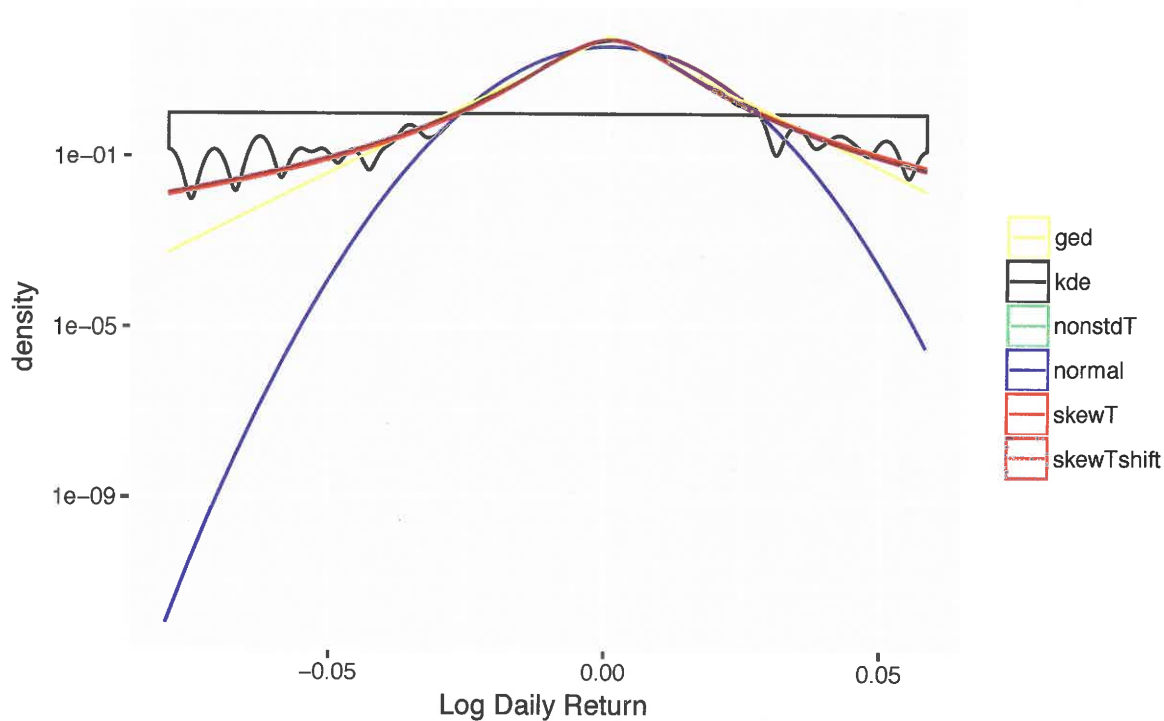


Data for Kellogg's (K)

January 2010 through December 2016



- 1 Compare all of these models by calculating the log likelihood for each:

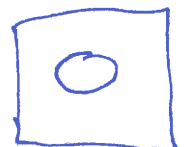
```
> print(sum(dged(ldrK$daily.returns, gedout$mle[1], gedout$mle[2],
+               gedout$mle[3], log=T)))
[1] 5770.239
```

```
> print(sum(dstd(ldrK$daily.returns, nonstdTout$mle[1],
+               nonstdTout$mle[2], nonstdTout$mle[3], log=T)))
[1] 5790.015
```

```
> print(sum(dSkewT(ldrK$daily.returns, skewtout$mle[1],
+                 skewtout$mle[2], skewtout$mle[3], skewtout$mle[4], log=T)))
[1] 5788.42
```

```
> print(sum(dSkewT(ldrK$daily.returns, skewtshiftout$mle[1],
+                 skewtshiftout$mle[2], skewtshiftout$mle[3],
+                 skewtshiftout$mle[4], skewtshiftout$mle[5], log=T)))
[1] 5790.34
```

```
> print(sum(dnorm(ldrK$daily.returns, normalout[1], normalout[2], log=T)))
[1] 5592.998
```



1 **Exercise:** Should these five models be compared using their likelihoods?

2 No. They are of varying complexity, i.e. some
3 have more flexibility than others.

4
5 For example, the GED includes the normal
6 as a special case, so the log likelihood for
7 the GED must be at least as large as
8 the log likelihood for normal

1 **Overfitting**

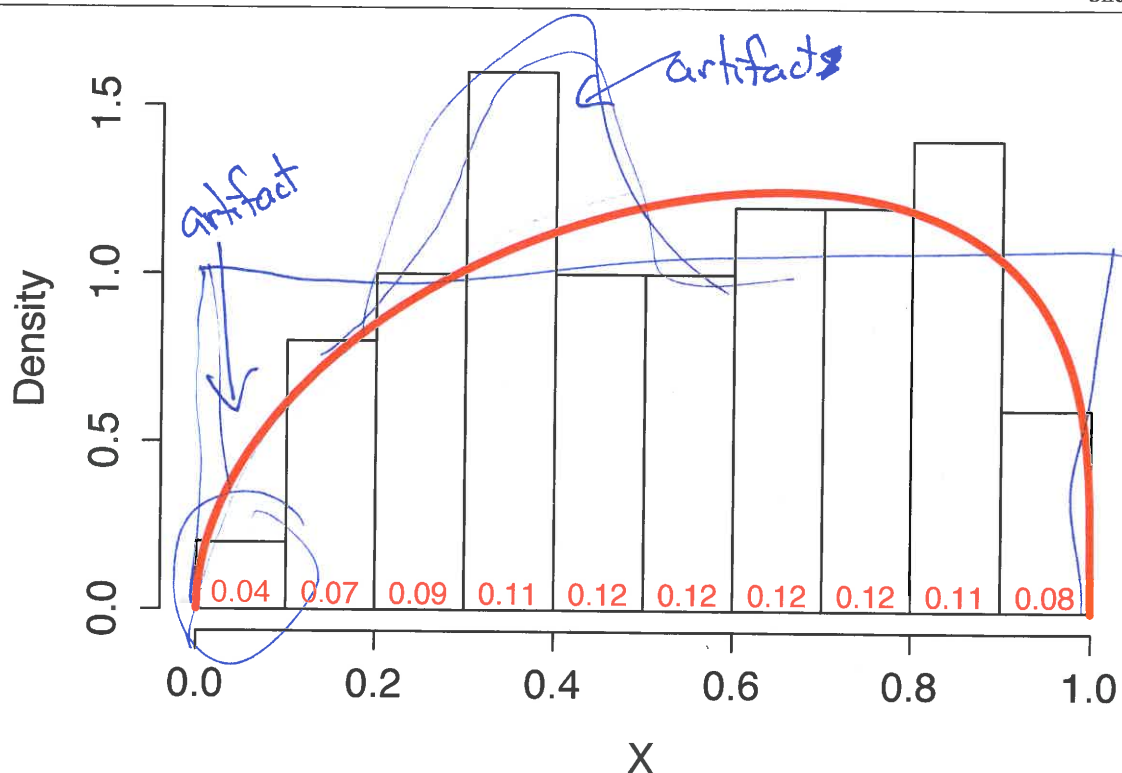
2 Model 1 is **nested** in Model 2 if Model 1 is a special case of Model 2. In this
3 case, the likelihood maximized under Model 2 will necessarily be greater
4 than or equal to the likelihood maximized under Model 1.

5 In our example, Normal is nested in the GED and the nonstandard t dis-
6 tribution is nested in the skewed, generalized t distribution with drift.

7 **But:** Increased likelihood may not be a sign of a better fitting model. It is
8 possible that we are simply **overfitting** to the available sample.

9 Our objective is to fit well to the **population**, not just to the observed sam-
10 ple. The sample may contain **artifacts** which appear to be real features, but
11 would disappear with a larger sample size.

- 1 **Example:** Suppose that a sample of size $n = 50$ is drawn from the
2 $\text{Uniform}(0, 1)$ distribution. If we inspect the histogram of this sample, it is
3 not going to be “flat” due to natural variability.
4 We may be tempted to fit a $\text{Beta}(\alpha, \beta)$ distribution to this sample, in an
5 effort to fit to any “features” we observe. This would be a mistake, and
6 would be overfitting.
7 See the figure on the following slide. Maximum likelihood is used to fit
8 the Beta distribution to the uniform data. The numbers in the bins show
9 the probabilities that the fitted Beta distribution assigns to each of those
10 intervals. Of course, the true probability is 0.10 for all ten bins.
11 **Exercise:** Label the “artifacts” in this histogram.



1 Models with more parameters have greater flexibility to fit to features, but
2 also artifacts, in observed data. Hence, the risk of overfitting increases
3 greatly as we consider models with a larger number of parameters (and
4 more complexity).

5 Akaike (1973) proposed a simple way to balance likelihood and model
6 complexity, and this has proven to be a very valuable tool for **model se-**
7 **lection**. Define

$$8 \quad AIC = -2 \log L(\theta) + 2p$$

9 where p is the number of parameters in the model. We seek the model that
10 has the **smallest** value of AIC.

11 Think of $2p$ as being a “penalty” on the likelihood for the complexity of the
12 model.

1 **Exercise:** Which of our five candidate models is selected by AIC in the
2 Kellogg’s example?

$$3 \quad \text{GED: } AIC = (-2)(5770.239) + (2)(3) = -11534.48$$

$$4 \quad \text{Nonstandard T: } p=3, AIC = -11574.03 \leftarrow$$

$$5 \quad \text{Skew T: } p=4, AIC = -11568.84$$

$$6 \quad \text{with shift: } p=5, AIC = -11570.68$$

$$7 \quad \text{Normal: } p=2, AIC = -11182$$

8
9 Based on AIC, nonstandard T is best choice

- 1 **Exercise:** How could AIC be utilized in the selection of predictors for a
2 regression model?

3 For any regression model, can calculate the
4 value of AIC. The "penalty" will
5 equal $2 \times$ predictors

6 As include more predictors, likelihood will definitely
7 increase (i.e. RSS will decrease), need to
8 penalize for # of predictors to avoid
9 overfitting
10

11

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad \epsilon_i \sim N(0, \sigma^2)$$

Part 15: Hypothesis Testing

Summary

Statistical hypothesis testing is a formal framework for “making discoveries.” It is ubiquitous in research, and overused. Nevertheless, it can be useful, and it is definitely important to understand the concept.

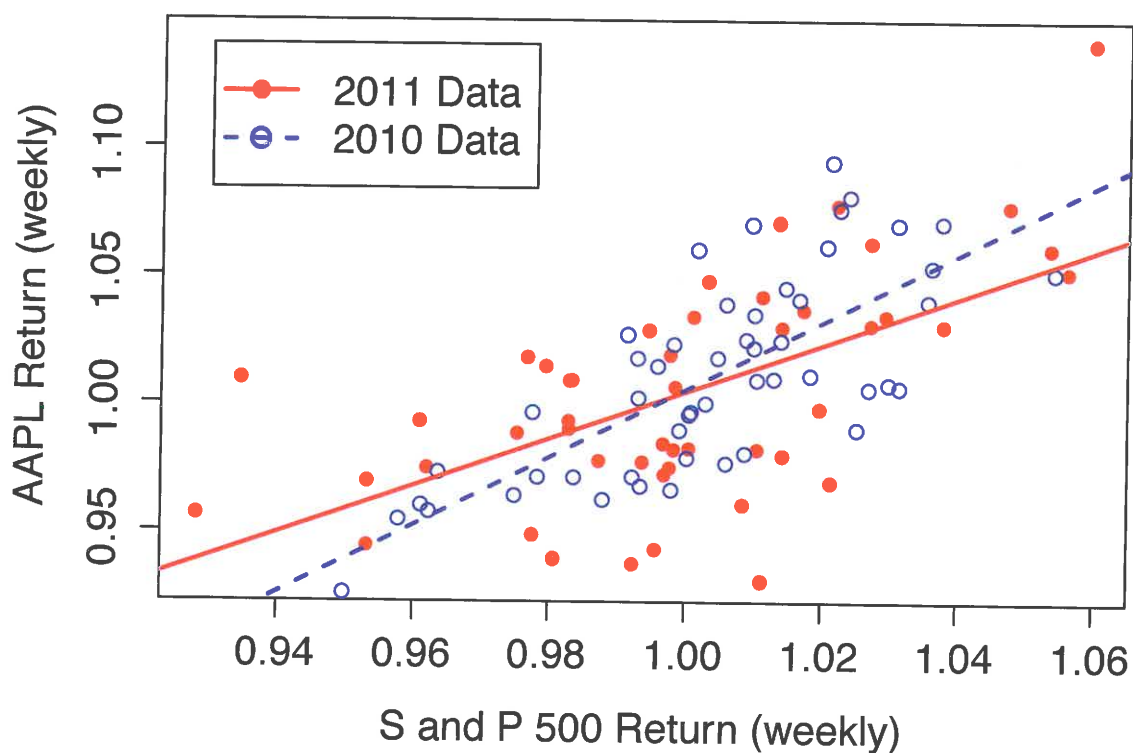
In particular, one will see repeated references to **p-values** when reading research. These are often misinterpreted.

Broadly speaking, hypothesis testing gives us a basis to make a claim that an observed “feature” is “real” and not simply an artifact that occurred due to random variability in our sample.

Consider the questions on the following three slides. In each case, the question is not asking for the best estimate of a parameter, but instead asking if the data provide sufficient evidence for some claim.

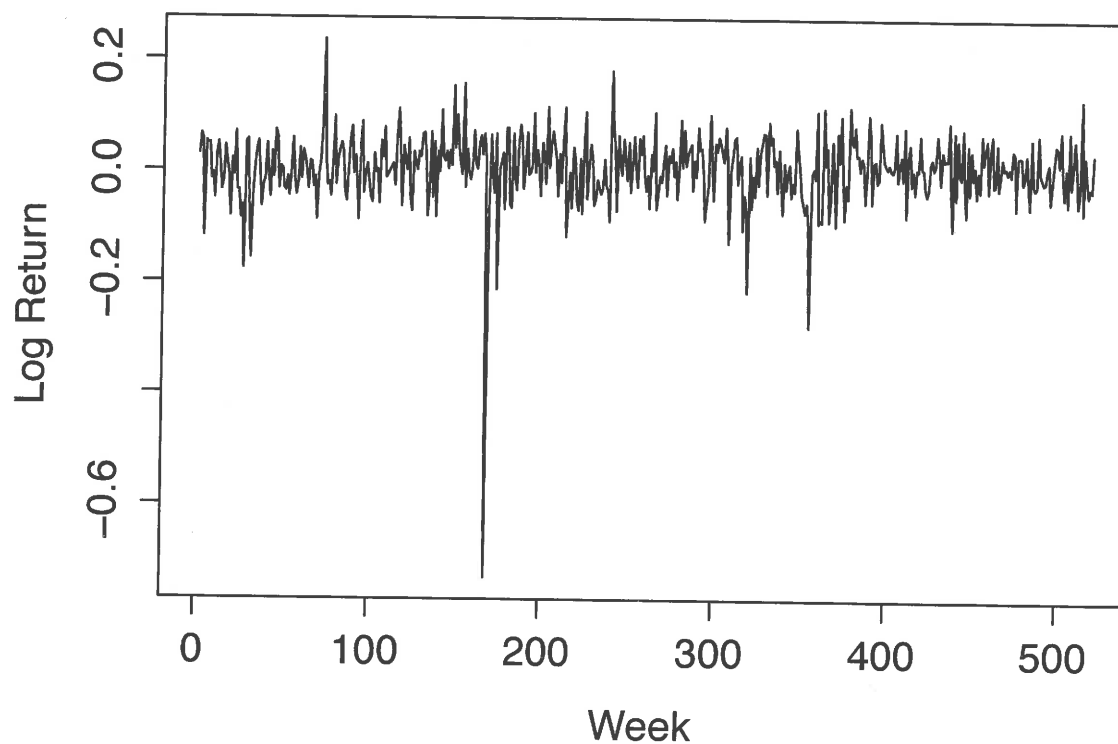
What do these examples have in common?

1. They each ask about a feature of the **population** from which the **sample** (observed data) is drawn.
2. The answer to each question is definitely “No.” The issue is whether or not the data provide **sufficient evidence** that the answer is “No.”
3. Each question puts forth a relatively simple “null hypothesis” and asks “Is there sufficient evidence to reject the null?”



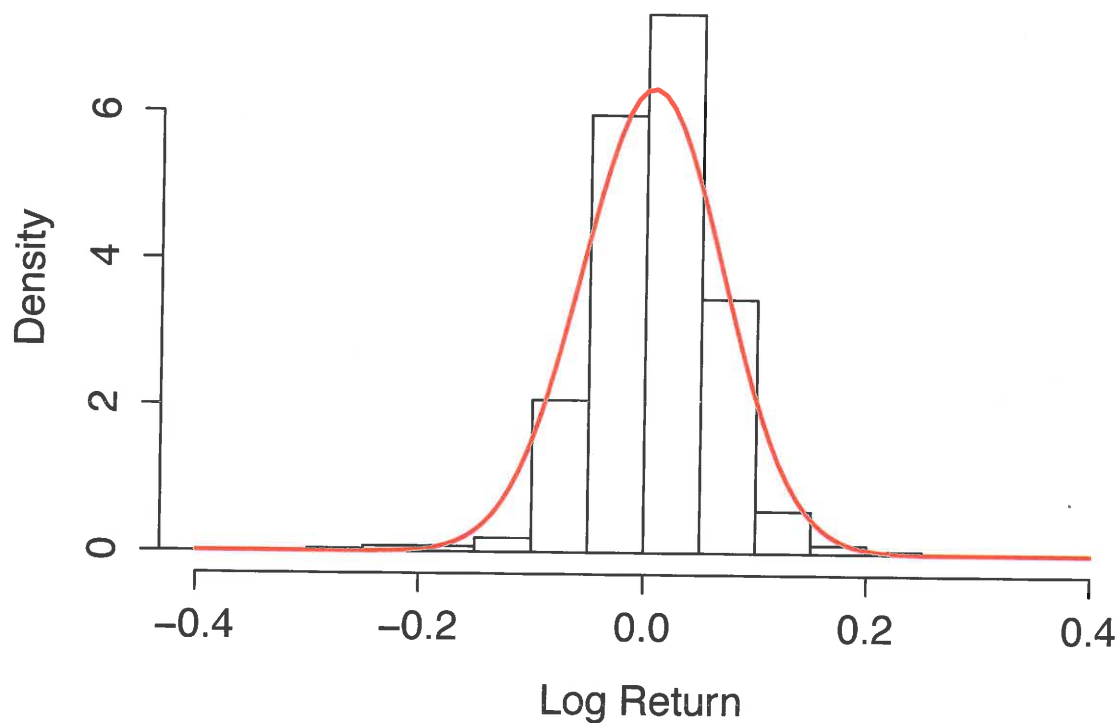
1

- 2 Is the slope of the regression line the same in 2010 and 2011?



1

- 2 Is the volatility constant over this time period?



1

2 Are these data normally distributed?

1 Such questions as these can be addressed using statistical hypothesis test-
2 ing.

3 The key requirement is that the question of interest, "Is there sufficient
4 evidence that...", can be turned into a question regarding the population.

5 Often the hypothesis will take the form of a question regarding θ , but this
6 is not always the case.

1 Null and Alternative Hypotheses

- 2 The first two key ingredients to a hypothesis test are the **null hypothesis**
3 (H_0) and the **alternative hypothesis** (H_a or H_1).
- 4 The alternative hypothesis is the claim you are testing to see if there is
5 strong enough evidence to support.
- 6 The null hypothesis is the "baseline" or "status quo."
- 7 The main thing to keep in mind is the following: **Statistical hypothesis**
8 **tests are designed to determine if there is strong evidence to support the**
9 **alternative (and reject the null). They are not designed to test if there is**
10 **strong evidence to support the null hypothesis.**

- 1 **Exercise:** A pharmaceutical company is testing a new treatment for a dis-
2 ease. The current treatment is 40% effective. Explain why a hypothesis
3 test is appropriate in this case. Formulate a reasonable null and alternative
4 hypothesis for the company.

5 Let p = effectiveness of new treatment

6

7 $H_0: p = 0.40$ vs. $H_1: p > 0.40$

8

9 Hyp. testing makes sense because company wants
10 strong evidence that $p > 0.40$ before taking
11 on the expense and risk of this new
12 treatment

- 1 **Exercise:** Write out null and alternative hypotheses for each of the three
 2 original examples. Be clear about what is being tested in each case.

3 ① $\beta_{1,2010}$ = slope in 2010

4 $\beta_{1,2011}$ = slope in 2011

5 $H_0: \beta_{1,2010} = \beta_{1,2011}$ vs. $H_1: \beta_{1,2010} \neq \beta_{1,2011}$

6 "Is there strong evidence that the slope has changed?"

7 ② $\sigma^2(t)$ = vol. at time t

8 $H_0: \sigma^2(t) = \sigma^2$ for all t vs. $H_1: \sigma^2(t)$ varies with t

9 "Is there strong evidence that the vol. is not constant?"

10 ③ H_0 : pop. is normal

11 H_1 : pop. is not normal

"Is there strong evidence that the pop. is not normal?"

1 The Test Statistic and the Rejection Region

- 2 Not surprisingly, data are used to test the hypothesis. As before, random
 3 variables $X_1, X_2, \dots, X_n$ are used to model the observable sample.
- 4 The **test statistic** is the function of the data that we will base our conclusion
 5 on. Keep in mind that the test statistic is also a random variable.
- 6 Write $T = T(X_1, X_2, \dots, X_n)$.
- 7 The **rejection region** is the set of values for the test statistic such that the
 8 conclusion is "reject the null hypothesis in favor of the alternative." In
 9 other words, when T falls into the rejection region, we conclude that there
 10 is strong evidence in support of the alternative.

- 1 How is the rejection region chosen?
- 2 Consider the two possible errors we can make with a hypothesis test:
 - 3 1. A **Type I error** is made when we reject the null even though the null is
 - 4 true.
 - 5 2. A **Type II error** is made when we fail to reject the null when the alter-
 - 6 native is true.
- 7 The assumption is that we are **primarily concerned with making a Type I**
- 8 **error**. We want to limit the probability of making a Type I error. ^{not}
- 9 We form the rejection region so that the probability of a Type I error is less
- 10 than or equal to some chosen amount α . This is often called the "size" or
- 11 "level" of the test.

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, with σ^2
- 2 known. We want to test $H_0: \mu = 10$ versus $H_1: \mu > 10$. A natural test
- 3 statistic in this case is $\bar{X}$. Formulate the rejection region for a level α test.

4 $T = \bar{X}$. The rejection region should be of

5 the form (c, ∞) for some $c > 10$.

6 How to choose c ?

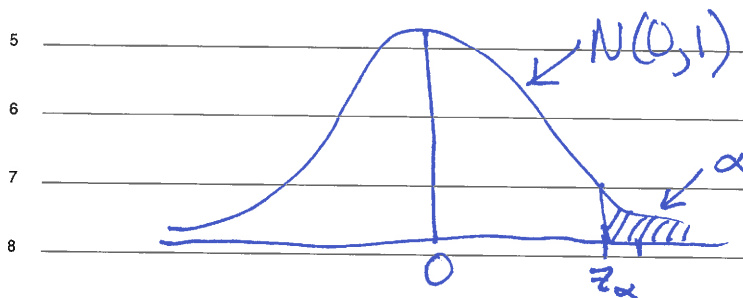
7 Fix some level α . Find c such that

$$8 \quad P(\text{Type I error}) \leq \alpha$$

$$9 \quad P(\bar{X} > c \mid H_0 \text{ is true}) \leq \alpha$$

$$P\left(\frac{\bar{X}-10}{\sigma/\sqrt{n}} > \frac{c-10}{\sigma/\sqrt{n}} \mid H_0 \text{ is true, i.e. } \mu=10\right) \leq \alpha$$

$$P\left(Z > \frac{c-10}{\sigma/\sqrt{n}}\right) \leq \alpha \quad \text{where } Z \sim N(0,1)$$



$$\text{Need } \frac{c-10}{\sigma/\sqrt{n}} \geq z_\alpha$$

$$\text{i.e. } c \geq \underbrace{10 + z_\alpha \sigma/\sqrt{n}}$$

In practice, find RR (the value of c)
so that $P(\text{Type I error}) = \alpha$

$$\text{In this example, } c = 10 + \underbrace{z_\alpha \left(\sigma/\sqrt{n}\right)}_{\text{tolerance}}$$

As $n \nearrow \infty$, tolerance decreases

As $\alpha \searrow 0$, $z_\alpha \nearrow$ so tolerance increases

1 In the previous exercise, we note that the result was based on knowledge
 2 of the distribution of the test statistic under the null hypothesis. It is key
 3 that the test statistic be chosen so that its distribution is known (at least
 4 approximately), under the null. As usual, the central limit theorem can be
 5 very helpful.

6 Also, we note that in the exercise we tested the **one-sided alternative hy-**
 7 **pothesis** $H_1: \mu > 10$. In other cases we will test a **two-sided alternative**, for
 8 example, $H_1: \mu \neq 10$. In such a case the null would be rejected for either
 9 small or large values of the test statistic.

1 **Exercise:** Suppose that we tested $H_1: \mu \neq 10$ in the above example. What
 2 would the rejection region be for fixed α ?

$$(-\infty, c_1) \cup (c_2, \infty) \quad \text{with } T = \bar{X}$$

$$c_1 < 10 < c_2$$

